

Midterm Exam

BSTA 551: Theory of Statistical Inference

Exam Instructions

This is a take-home exam. You have approximately 2 weeks to complete it.

- **Time estimate:** This exam is designed to take approximately 1.5 hours.
- **Resources:** You may use your notes, textbook (Devore, Berk, Carlton), and R/RStudio. You may NOT consult with other students or use AI assistants.
- **Submission:** Submit your work as a PDF (rendered from Quarto/RMarkdown) or a single PDF of handwritten work. Include all R code with `echo = TRUE`.
- **Show your work:** Full credit requires complete derivations and explanations. For computational problems, include your R code.
- **Seed:** For all simulation problems, use `set.seed(551)` for reproducibility.
- **Academic integrity:** By submitting this exam, you affirm that all work is your own.

Point values are indicated for each problem. Total: 100 points.

Part I: Conceptual Questions (45 points)

Problem 1: Estimator Properties (15 points)

A pharmaceutical company is developing a new assay to measure drug concentration in blood plasma. Let $X_1, X_2, \dots, X_n$ be independent measurements from a distribution with mean μ (the true concentration) and variance σ^2 .

Three estimators for μ are proposed:

- $\hat{\mu}_1 = \bar{X}$ (sample mean)
- $\hat{\mu}_2 = \frac{X_1 + X_n}{2}$ (average of first and last observations)
- $\hat{\mu}_3 = \frac{1}{n+1} \sum_{i=1}^n X_i$ (weighted sample mean)

(a) (3 points) For each estimator, determine whether it is unbiased for μ . If biased, calculate the bias.

(b) (4 points) Calculate the variance of each estimator in terms of σ^2 and n . Assume the observations are independent.

(c) (4 points) Calculate the Mean Squared Error (MSE) of each estimator. Which estimator has the smallest MSE for $n = 10$?

(d) (4 points) As $n \rightarrow \infty$, which estimator(s) are consistent for μ ? Justify your answer using the definition of consistency.

Problem 2: Fisher Information and the Cramér-Rao Lower Bound (15 points)

A biostatistician models the total time (in weeks) for a patient to complete a two-phase post-operative rehabilitation protocol after hip replacement surgery. Each phase has an independent, identically distributed duration, so the total rehabilitation time follows a Gamma distribution with known shape $\alpha = 2$ and unknown rate parameter $\beta > 0$. The pdf is:

$$f(x; \beta) = \beta^2 x e^{-\beta x}, \quad x > 0$$

For this distribution, $E(X) = 2/\beta$ and $\text{Var}(X) = 2/\beta^2$.

(a) (4 points) Calculate the Fisher Information $I(\beta)$ for a single observation. Show your derivation using $I(\beta) = E \left[-\frac{d^2}{d\beta^2} \ell(\beta) \right]$.

(b) (3 points) State the Cramér-Rao Lower Bound (CRLB) for unbiased estimators of β based on n independent observations.

(c) (4 points) The MLE of β can be shown to be $\hat{\beta}_{MLE} = 2/\bar{X}$. It can also be shown that $\sum_{i=1}^n X_i \sim \text{Gamma}(2n, \beta)$, and for $Y \sim \text{Gamma}(a, b)$, we have $E(1/Y) = b/(a-1)$ when $a > 1$.

- Use these facts to show that $\hat{\beta}_{MLE}$ is biased, and find the bias.
- Propose a modified estimator $\tilde{\beta}$ that is unbiased for β .

(d) (4 points) It can be shown that for $Y \sim \text{Gamma}(a, b)$, $\text{Var}(1/Y) = \frac{b^2}{(a-1)^2(a-2)}$ when $a > 2$. Use this to find $\text{Var}(\tilde{\beta})$ and determine whether the unbiased estimator achieves the CRLB. What does this result tell us?

Problem 3: Confidence Interval Interpretation (15 points)

A clinical trial investigates the effectiveness of a new statin drug in lowering LDL cholesterol. In a sample of $n = 64$ patients, the mean reduction in LDL cholesterol is $\bar{x} = 28$ mg/dL with sample standard deviation $s = 16$ mg/dL.

(a) (4 points) Construct a 95% confidence interval for the true mean reduction μ . Justify your choice of interval (z vs. t) and state any assumptions.

(b) (4 points) For each statement below, indicate whether it is a CORRECT or INCORRECT interpretation of your confidence interval. Briefly explain why.

- i. "There is a 95% probability that μ lies in this interval."
- ii. "If we repeated this study many times, 95% of the resulting confidence intervals would contain μ ."
- iii. "95% of patients in this study had LDL reductions within this interval."
- iv. "We are 95% confident that the true mean LDL reduction is in this interval."

(c) (4 points) The drug company wants to claim that the mean LDL reduction is at least 25 mg/dL. Based on your confidence interval, can they make this claim at the 95% confidence level? Explain.

(d) (3 points) If the company wanted a 95% CI with margin of error no greater than 3 mg/dL, what sample size would be required? (Use $s = 16$ as your estimate of σ .)

Part II: Computational Problems (55 points)

Problem 4: Maximum Likelihood Estimation (18 points)

The number of emergency room visits per month at a rural hospital follows a Poisson distribution with unknown rate μ . Over 12 months, the following counts were recorded:

$$X = \{47, 53, 61, 42, 58, 55, 49, 64, 52, 45, 59, 51\}$$

(a) (4 points) Write the likelihood function $L(\mu)$ and log-likelihood function $\ell(\mu)$ for the observed data.

(b) (4 points) Derive the MLE $\hat{\mu}$ of μ by maximizing the log-likelihood. Calculate the numerical value of $\hat{\mu}$ for this data.

(c) (4 points) Using the Neyman Factorization Theorem, show that $T = \sum_{i=1}^n X_i$ is a sufficient statistic for μ .

(d) (6 points) Using R:

- Calculate the MLE $\hat{\mu}$
- Calculate the standard error of $\hat{\mu}$ using the observed Fisher Information
- Construct an approximate 95% Wald confidence interval for μ

```
# ER visit data
er_visits <- c(47, 53, 61, 42, 58, 55, 49, 64, 52, 45, 59, 51)

# Your code here
```

Problem 5: Method of Moments vs. MLE (15 points)

The survival time (in years) of patients after a certain cardiac procedure is modeled using a Weibull distribution with shape parameter $k = 2$ (known) and unknown scale parameter $\lambda > 0$. The pdf is:

$$f(x; \lambda) = \frac{2x}{\lambda^2} \exp\left(-\frac{x^2}{\lambda^2}\right), \quad x > 0$$

For this distribution: $E(X) = \lambda \cdot \frac{\sqrt{\pi}}{2}$ and $E(X^2) = \lambda^2$

The following survival times (in years) were observed for $n = 10$ patients:

$$X = \{3.2, 4.8, 2.1, 5.5, 3.9, 4.2, 6.1, 3.5, 4.6, 2.8\}$$

(a) (5 points) Derive the Method of Moments estimator $\tilde{\lambda}$ for λ using the first moment. Calculate its numerical value for this data.

(b) (5 points) Derive the MLE $\hat{\lambda}$ for λ . Calculate its numerical value for this data.

(c) (5 points) Using R simulation with 5000 replications, compare the bias, variance, and MSE of the two estimators assuming the true parameter is $\lambda = 4$. Which estimator performs better? Use $n = 10$.

```
set.seed(551)

# Simulation parameters
n <- 10
lambda_true <- 4
n_sims <- 5000

# Your simulation code here
```

Problem 6: Confidence Intervals for Proportions (12 points)

In a vaccine efficacy trial, 312 out of 400 vaccinated individuals developed protective antibody levels after receiving a new COVID-19 booster.

(a) (3 points) Calculate the sample proportion $\hat{p}$ and its standard error (using the Wald formula).

(b) (4 points) Compute three different 95% confidence intervals for the true proportion p :

- Wald interval
- Score (Wilson) interval (use `prop.test()` in R)
- Agresti-Coull interval

(c) (3 points) Which interval would you recommend for reporting in a medical journal, and why?

(d) (2 points) The FDA requires that the lower bound of a 95% CI for vaccine efficacy exceed 70% for approval. Based on your preferred interval, would this vaccine meet the FDA threshold?

```
# Vaccine trial data
x <- 312 # successes
n <- 400 # total

# Your code here
```

Problem 7: Bootstrap Confidence Intervals (10 points)

Researchers measured the time-to-recovery (in days) for $n = 15$ patients receiving a new physical therapy protocol after knee surgery:

$$X = \{18, 22, 25, 19, 31, 24, 20, 28, 23, 45, 21, 26, 24, 22, 27\}$$

Notice the potential outlier at 45 days.

(a) (2 points) Calculate the sample mean, sample median, and sample standard deviation. Create a histogram or Q-Q plot to assess normality.

(b) (4 points) Using $B = 10,000$ bootstrap replications, compute:

- The bootstrap standard error of the sample mean
- The 95% bootstrap percentile confidence interval for the population mean
- The 95% bootstrap percentile confidence interval for the population median

(c) (2 points) Calculate the classical t-interval for the mean. How does it compare to the bootstrap percentile interval?

(d) (2 points) For this data, would you recommend using the mean or median to summarize the center of the distribution? Which confidence interval (bootstrap or classical) would you report? Justify your answers.

```
set.seed(551)

# Recovery time data
recovery <- c(18, 22, 25, 19, 31, 24, 20, 28, 23, 45, 21, 26, 24, 22, 27)

# Your code here
```


Bonus Question (5 points)

(Optional - for extra credit)

A biostatistician is estimating the prevalence of a rare genetic mutation. In a sample of $n = 500$ individuals, only $X = 3$ carry the mutation.

(a) (2 points) Calculate the Wald, Score, and Agresti-Coull 95% confidence intervals. What problem do you observe with the Wald interval?

(b) (3 points) Explain why the Score and Agresti-Coull intervals are more appropriate for rare events. What property of these intervals addresses the limitation of the Wald interval?

```
# Rare mutation data
x <- 3
n <- 500

# Your code here
```

End of Exam

Good luck!